

Retrieval-Augmented Generation for Enterprise Knowledge Systems

Abstract

Large language models are powerful general-purpose reasoners, but they are prone to hallucination when asked about domain-specific or proprietary information that was not present in their training data. Retrieval-Augmented Generation (RAG) addresses this limitation by retrieving relevant passages from an external knowledge base at query time and conditioning the model's response on that retrieved context. This paper describes a production RAG architecture built around dense vector retrieval, hybrid search, and citation-grounded generation.

1. Methodology

Our pipeline ingests PDF documents, extracts text with page-level metadata, and splits the text into overlapping chunks of approximately 1000 characters with a 150 character overlap. Each chunk is embedded using a sentence transformer model and stored in a vector database alongside its source document and page number. At query time, the system supports three retrieval modes: semantic (dense vector similarity), keyword (sparse term matching), and hybrid, which fuses both signals. Retrieved chunks are inserted into a structured prompt that instructs the language model to answer strictly from the provided context and to decline to answer when the context is insufficient.

2. Advantages and Limitations

The primary advantage of this approach is grounding: every answer can be traced back to a specific document and page, which builds user trust and supports auditability. A further advantage is that the knowledge base can be updated without retraining the underlying language model. A limitation is that retrieval quality bounds answer quality; if the relevant passage is not retrieved, the model cannot answer correctly even if it 'knows' the fact from pretraining. Chunking strategy therefore has an outsized effect on system performance.

3. Conclusion

We conclude that hybrid retrieval combined with strict context-grounding and explicit citations is an effective architecture for enterprise question answering over private document collections, and that conversation memory is necessary to correctly resolve follow-up questions that refer back to earlier turns in a session.